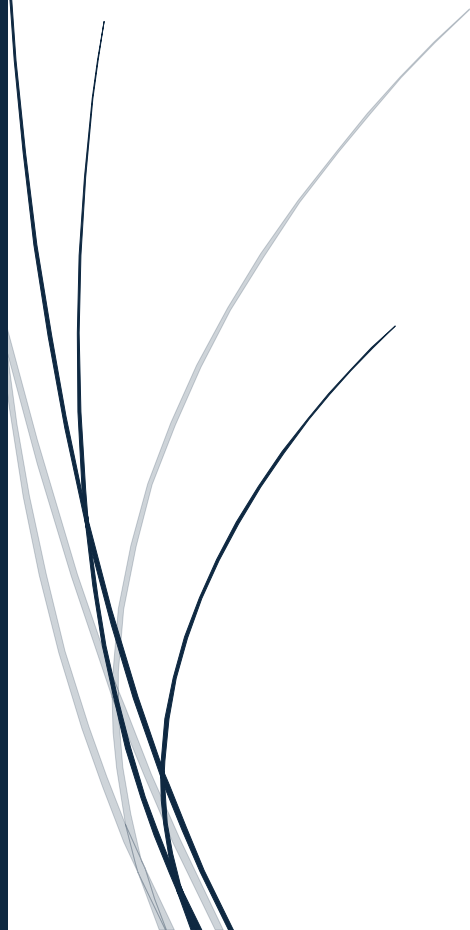




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# Analyse of ransomware

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## Abstract

Ransomware is malware that encrypts sensitive data on a computer system, making it inaccessible until a ransom is paid.

This growing threat in the digital environment is jeopardising data security and the stability of IT systems.

The aim of this report is to explore the subject of ransomware, highlighting its impact on IT security and the importance of detailed analysis to prevent and mitigate these attacks. We provide recommendations for improving your IT security by examining the fundamentals of software security, an in-depth analysis of ransomware, analysis methods, lessons learned from this analysis and prospects for integration into software development. Collaboration and knowledge sharing within the security community are also essential to effectively combat ransomware.

## Introduction

Ransomware is a growing threat in today's digital environment, jeopardising data security and the stability of IT systems.

Given this problem, it is important to understand exactly how this malware works to deal with it more effectively.

This report aims to explore the issue of ransomware, highlight its impact on IT security and emphasise the importance of in-depth analysis to prevent and mitigate these attacks.

The report covers the fundamentals of software security, in-depth analysis of ransomware, analysis methodologies, lessons learned from this analysis, prospects for integration into software development and the importance of collaboration and knowledge sharing in the fight against ransomware.

# Foundations of Software Security

Software security is based on four fundamental principles: confidentiality, integrity, availability, and authentication/authorisation. Ensuring data confidentiality, maintaining data integrity, guaranteeing availability, and controlling user access are essential to protect systems against attacks such as ransomware.

The most common software vulnerabilities include unpatched vulnerabilities, programming errors, misconfigurations, and missing access controls.

These vulnerabilities can be exploited by an attacker to compromise your system. By understanding these principles and applying security best practice, developers and administrators can reduce the risk of attacks such as ransomware.

Regular maintenance, secure system configuration and constant vigilance are essential to maintaining a secure environment.

## In-depth analysis of ransomware

Ransomware is a growing threat to IT security, and a detailed analysis of its inner workings is essential to better understand its mechanisms and develop effective countermeasures. Under the bonnet of ransomware is designed to encrypt sensitive data on infected systems and prevent legitimate users from accessing the files.

A typical ransomware workflow involves several important steps.

**Initial infection:** Ransomware enters target systems via attack vectors such as phishing emails, compromised websites or software vulnerabilities.

**Trigger:** Once the ransomware becomes active on a target system, it begins encrypting files using a powerful, mainly asymmetric, encryption algorithm. With this algorithm, only a private key held by the attacker allows decryption.

**Ransom notes:** After encrypting the data, the ransomware displays a ransom note (usually in the form of a ransom demand) to the user and asks them to pay the ransom in exchange for the decryption key.

**Encryption for data encryption** Ransomware generally encrypts data using asymmetric encryption algorithms such as RSA or ECC. These algorithms require a pair of keys: a public key for encryption and a corresponding private key for decryption. Ransomware attacks generate a unique key pair for each infection, send the public key to the target system and store the private key on a remote server controlled by the attacker. “The findings of the research indicate that responses to crypto-ransomware are made more complex by the nuanced relationship between the technical (malware which encrypts) and the human (social engineering which still instigates most infections) aspects of an attack.” (Lena Y. Connolly, 2019)

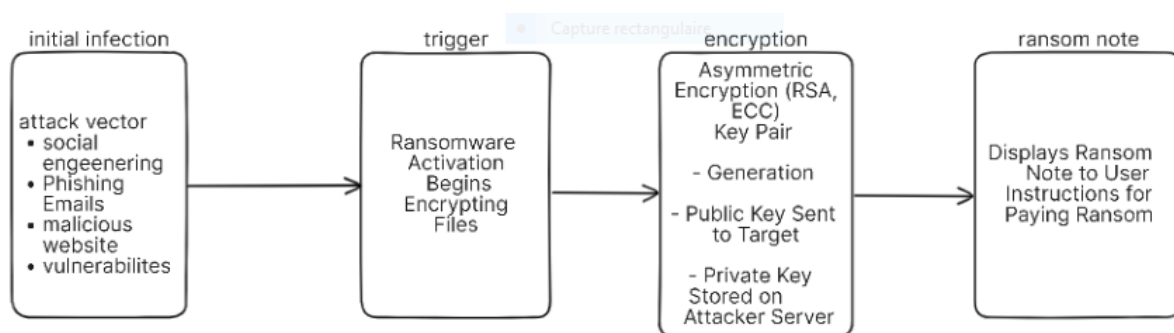


Figure 1 Ransomware Infection Process

## Analysis methodologies

Ransomware analysis requires a combination of specialist techniques and tools to understand exactly how this malware works. This section describes common methods of ransomware analysis and highlights the importance of collaboration within the security community. “The Operational Design methodology of military forces helps in better understanding how APTs devise their strategies.” (Sun., 2023)

### Tools and techniques used in ransomware analysis

**Sandbox:** A sandbox is an isolated environment in which ransomware can be run safely without risk of contaminating the host system. This allows you to observe the behaviour of ransomware, understand the actions it is trying to perform and capture indicators of compromise (IoCs).

Example tool: Cuckoo Sandbox

**Disassembly and decompilation:** Disassembly converts the ransomware's binary code into readable assembly code, while decompilation attempts to restore the original source code. These techniques are essential for analysing the malicious code and identifying key features such as encryption routines and evasion mechanisms.

Examples of tools: IDA Pro, Ghidra

**Dynamic analysis** Dynamic analysis involves running ransomware in a controlled environment and observing its behaviour in real time. This includes intercepting network communications, monitoring file system modifications and analysing process activity.

Examples of tools: Process Monitor

**Static analysis:** Static analysis is performed without executing malicious code. This includes inspecting binaries, looking for suspicious strings and identifying code signatures. This method is often used for an initial assessment before a more detailed dynamic analysis.

Examples of tools: String, PEiD

**Network and traffic monitoring** Ransomware communicates with command and control (C&C) servers to receive instructions and send data. Network traffic monitoring can help you capture and analyse these communications, providing important clues about which servers are being used and what data is being exposed.

Examples of tools include Wireshark, tcpdump

## The importance of collaboration in the security community

**Information sharing:** Platforms such as VirusTotal, security forums and working group specialists allow researchers to share their findings and collaborate on research.

Detailed analysis.

**Collaborative initiatives:** Initiatives such as the No More Ransom project, which brings together public and private partners, aim to provide free decryption tools and raise public awareness of the dangers of ransomware.

**Conferences and publications:** Security conferences and trade publications such as Black Hat and DEF CON provide an opportunity to present research findings, discuss emerging trends and develop collaborative strategies for protection against ransomware.

Advanced analytical tools and techniques, combined with close collaboration within the security community, can improve our ability to detect, analyse and respond to ransomware. This requires constant vigilance, regular updating of knowledge and a willingness to share information to strengthen collective security.

## Lessons Learned from Analysis

A detailed analysis of ransomware provides important information for strengthening IT security and developing effective defence strategies.

This section presents the key findings of our ransomware analysis, focusing on the identification of specific vulnerabilities and recommended mitigation measures.

### Identification of specific security vulnerabilities

**Operating system vulnerabilities:** Analysis of ransomware often reveals unpatched operating system vulnerabilities such as: B. Zero Day exploits or known but unpatched vulnerabilities. Ransomware exploits these vulnerabilities to gain initial access to the system or to increase privileges on the system.

Example: WannaCry exploited a vulnerability in the Windows SMB protocol (MS17-010).

**Application vulnerabilities:** Poorly secured or obsolete applications are prime targets for ransomware. Exploitable vulnerabilities can exist in browser plugins, third-party software, and even in-house applications.

Example: Malicious macros in Microsoft Office documents used by Locky.

**Security policy vulnerabilities:** Inappropriate security policies, such as insecure default settings or excessive privileges granted to users, increase a system's vulnerability to ransomware.

Example: Users with unnecessary administrative privileges facilitate the propagation of ransomware and the encryption of sensitive data.

## Recommended corrective actions.

**Reinforcement of security protocols:** The strict application of security protocols, such as patch management, regular software updates and disabling unnecessary services, is essential to reduce attack vectors.

Recommendation: Establish a patch management system to ensure that all known vulnerabilities are patched in a timely manner.

**User training and awareness:** Educating users about ransomware threats and social engineering techniques is important to prevent initial infections. Regular training programmes can help users recognise phishing attempts and adopt safe practices.

Recommendation: Provide training on recognising phishing emails and the importance of password security.

**Use advanced security solutions:** Deploying advanced security solutions such as anti-virus, anti-ransomware software and intrusion detection systems (IDS) can help detect and block ransomware before it can cause damage.

Recommendation: Deploy advanced security solutions that detect suspicious behaviour and protect against ransomware in real time.

**Regular, secure backups:** Implementing regular, secure backups of critical data is an essential measure to protect against ransomware. Backups should be stored offline or in a secure location to prevent any compromise by ransomware.

Recommendation: Implement a robust backup strategy with frequent backups and regular recovery tests to ensure data availability even in the event of an attack.

**Network segmentation:** Network segmentation can limit the spread of ransomware by isolating critical systems and limiting unauthorised communications between network segments.

Recommendation: Divide your network into logical segments and apply strict access controls and security policies tailored to each segment.

Analysis of ransomware provides valuable information for understanding its attack mechanisms and developing effective countermeasures. By identifying specific security vulnerabilities and implementing rigorous corrective measures, organisations can strengthen their resilience to ransomware.

# Prospects for Integration into Software

**Integrate security from the design stage Principle:** Integrate security measures from the earliest stages of development.

**Recommendation:** Use secure coding practices and carry out regular security analyses.

**Developer best practice Training:** Make developers aware of common security vulnerabilities and techniques.

**Tools:** Use static and dynamic code analysis tools to identify and correct security vulnerabilities.

By developing security by design and adopting best practice, developers can create software that is more resistant to ransomware and other threats.

## Development Collaboration and Knowledge Sharing

The fight against ransomware requires close collaboration and ongoing knowledge sharing between the various players in the cybersecurity field. This section highlights the importance of such cooperation and the various ways in which collective security can be strengthened.

**The importance of collaboration between security researchers:** Security researchers play an essential role in identifying new threats and developing effective countermeasures.

With your help, we can work together and benefit from a wide range of expertise in the fight against ransomware.

**Information sharing:** researchers share indicators of compromise (IoCs), malware samples and detailed analyses via a dedicated platform. This sharing enables faster, more coordinated responses to new threats.

**Examples:** VirusTotal, security forums, alerts.

**Community contributions to threat prevention:** An informed and engaged community is essential to the creation of a robust cybersecurity environment. Contributions to this community take many forms.

**Community projects:** Initiatives such as the No More Ransom project. Public and private sector partners come together to provide free decryption tools and raise awareness.

**Contributions to code and research:** researchers and security experts regularly publish analyses, tools, and recommendations for defending against ransomware.

**Conferences and seminars:** events such as Black Hat, DEFCON and other security conferences are important opportunities to share ideas, discuss emerging trends and collaborate on innovative solutions.

**Benefits of collaboration and sharing:** Increased responsiveness: by quickly sharing information about new threats, organisations can react more effectively.

**Develop common solutions:** Collaboration accelerates the development of common defence tools and techniques, reducing the time and cost of fighting ransomware.



**Strengthen collective capabilities:** Pooling knowledge and resources increases overall resilience to cyber threats.

Collaboration and knowledge sharing are fundamental pillars in the fight against ransomware. By working together and sharing information, the cyber security community can better prepare for and respond to attacks, and better protect sensitive systems and data against ransomware.

## Conclusion

In summary, ransomware analysis has proven to be an essential tool in the fight against cyber threats and in maintaining data security.

By exploring the inner workings of ransomware, identifying attack vectors, and providing concrete recommendations, we have been able to highlight the importance of vigilance and cooperation within the security community.

By integrating the information from this analysis into your software development practices and reinforcing security measures at the design stage, you can create systems that are more resistant to ransomware and emerging threats.

The future of software security depends on staying informed, sharing knowledge, fighting ransomware attacks and being proactive to ensure sensitive data is protected.

## Figures

Figure 1 Ransomware Infection Process ..... **Erreur ! Signet non défini.**

## Références

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